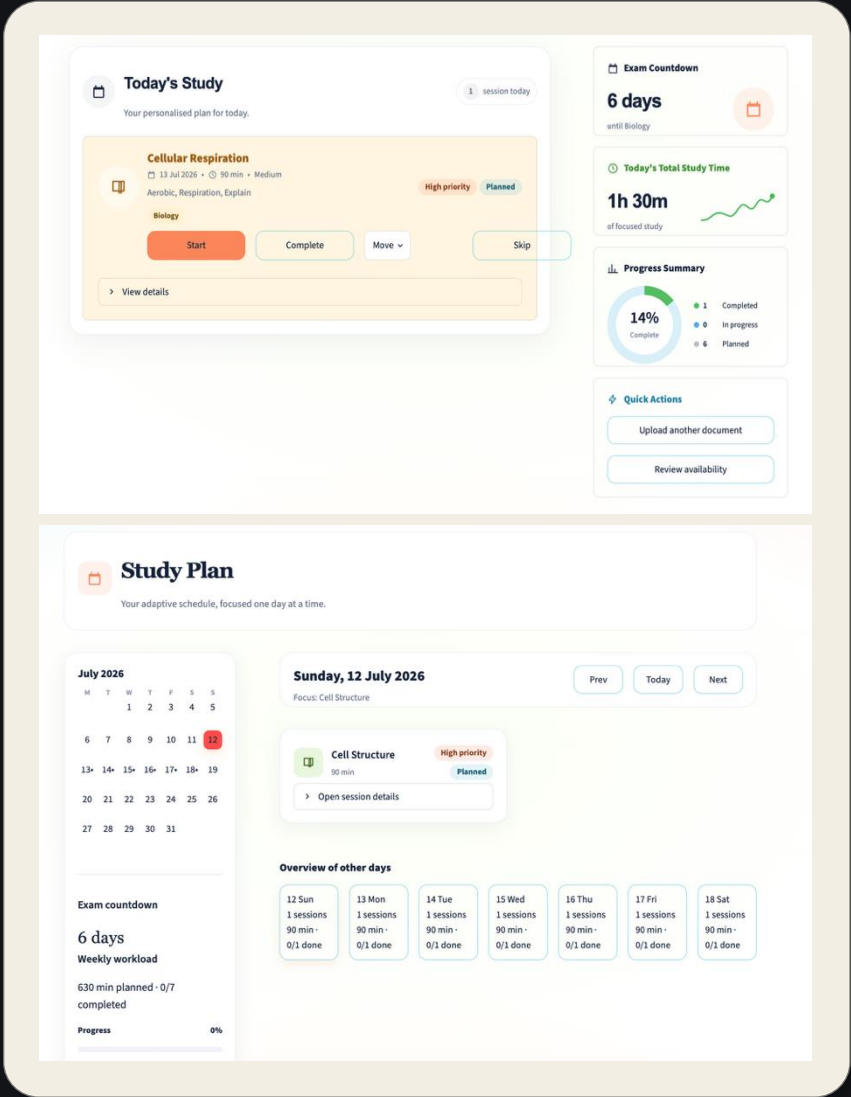


Rebound

The Plan After the Plan Fails

An adaptive exam-recovery planner that turns revision notes into a mastery-aware study plan — powered by AMD ROCm semantic compute.



Static study plans break the moment real life hits.

Students have notes, calendars and AI tutors. What they lack is a plan that understands topic relationships and recovers when time disappears.

01

Static plans break

One missed session cascades into the rest of the schedule.

02

No semantic understanding

Tools list topics but do not map how ideas depend on each other.

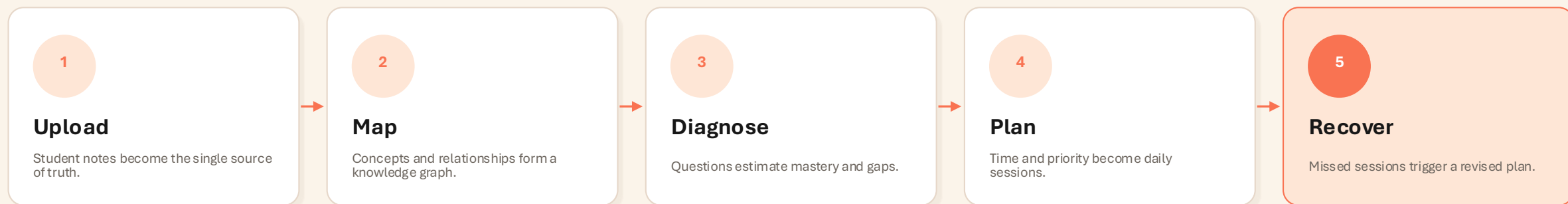
03

No recovery logic

Students must manually rebuild the plan when reality changes.

Upload. Map. Diagnose. Plan. Recover.

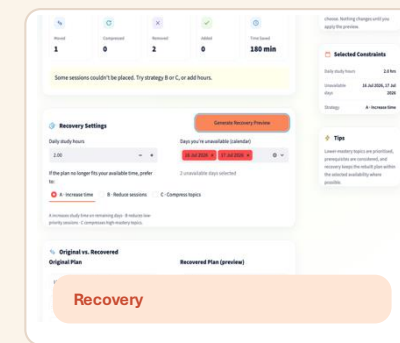
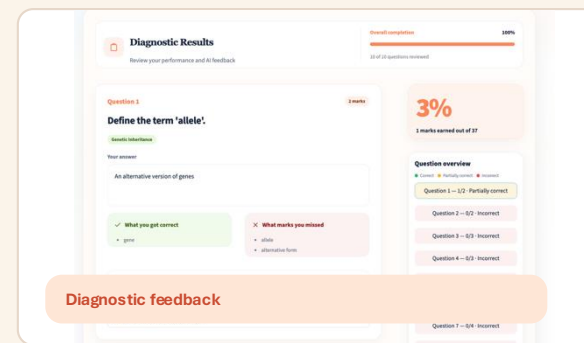
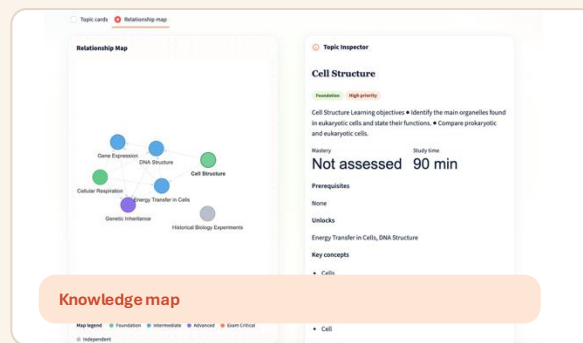
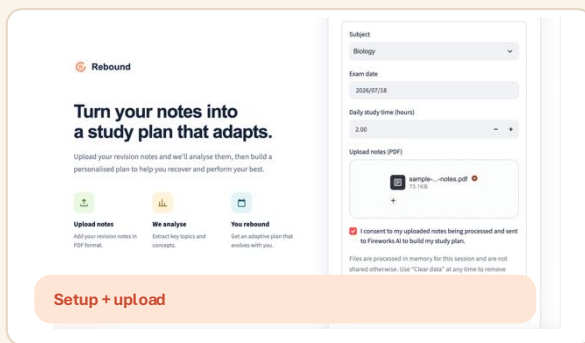
Five steps from raw notes to a resilient, personalised study plan.



Rebound turns notes into a plan that adapts as the student's mastery, available time, and schedule change.

From notes to plan in one session.

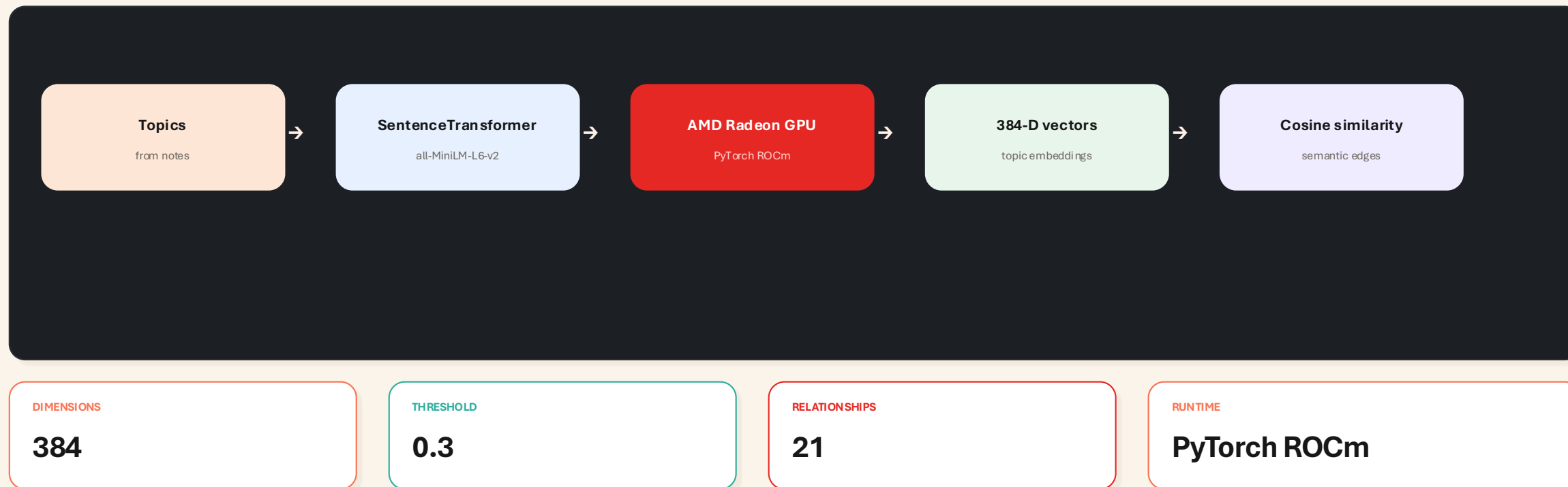
Screens from the live Rebound walkthrough: setup, knowledge map, diagnostic feedback and recovery.



Each recommendation is grounded in uploaded material, diagnostic evidence, available time and topic dependencies.

AMD ROCm is Rebound's semantic-compute layer.

An AMD Radeon GPU running PyTorch ROCm converts extracted topics into embeddings and discovers semantic relationships.



Reproducible AMD evidence, not a demo trick.

Every GPU run produces artifacts that the Streamlit app consumes and verifies at runtime.

embeddings.npy

Float32 matrix · (7, 384)

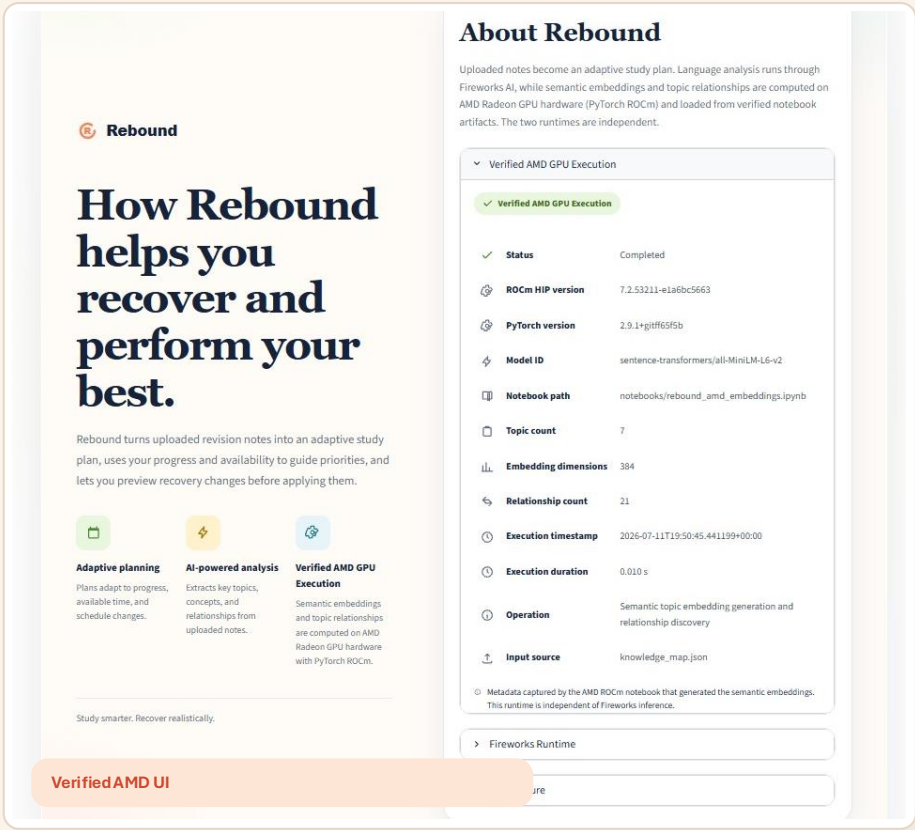
topic_metadata.json

Topic labels + relationships

amd_run_manifest.json

Device, ROCm, model, counts

✓ Verified AMD GPU Execution appears only when the manifest confirms a completed run.



Two runtimes. One coherent pipeline.

Rebound separates language reasoning from semantic compute so each runtime does what it does best.

Fireworks AI

Language reasoning

- Extracts topics and concepts
- Generates diagnostic questions
- Produces structured language output

AMD ROCm

Semantic compute

- Runs embedding model on Radeon GPU
- Calculates cosine similarity
- Exports reusable artifacts

[Student Notes](#) → [Fireworks Analysis](#) → [Knowledge Map](#) → [AMD Embeddings](#) → [Recovery Plan](#) → [Dashboard](#)

A bottom-up model built on verified demand signals.

Rebound targets students studying under deadline pressure, then expands into institutions and tutoring providers.

TAM

\$8.0B

222M tertiary students × \$36/year

SAM

\$0.8B

10% digital-first exam segment × \$36/year

SOM TARGET

\$3.6M

100K paying learners · internal target

Assumptions are labelled: 10% SAM segment, \$36/year pricing and 100K learner SOM are planning assumptions, not current performance.

Proposed pricing and competitive positioning.

Rebound sits between study-material tools, diagnostics and calendar rescheduling.

Free

1 active subject
basic diagnostic

Pro • ~\$3/mo

Full features
billed annually

Institution

Per-student
licensing

Future

LMS + white-label
planning API

Capability	Quizlet	StudyFetch	Motion	Rebound
Content understanding	Partial	Yes	No	Yes
Diagnostic mastery	No	Partial	No	Yes
Semantic topic mapping	No	No	No	Yes
Missed-session recovery	No	No	General	Exam-focused

From verified MVP to adaptive learning infrastructure.

Phase 1

Multi-subject support
Enhanced diagnostics

Phase 2

LMS integration
Institution pilot program

Phase 3

White-label planning API
EdTech partner integrations

Phase 4

Mobile app
Offline AMD inference

The Plan After the Plan Fails.

Rebound turns disruption into recoverable progress by combining language reasoning, AMD semantic compute and adaptive planning.



GitHub



Live demo

Team: Loh Yee Zhi · Khoo Yan She

github.com/cyeezz/rebound-amd